

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-1307
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	ServiceNow Platform UI (Next Experience / Core UI), HTML, CSS, Client Scripts, UI Policies	Provides a ready-made, responsive UI for forms and lists. Users see only the records they are permitted to access, so no separate front-end build is needed. Client Scripts and UI Policies handle the field-level display behaviour.
2	Backend / Server-Side	ServiceNow server-side JavaScript (Glide API: GlideRecord, GlideSystemUser), Script-based ACLs	ACL scripts evaluate a field value (e.g. record state, priority, department) and return true or false to allow or deny access. Enforcement happens on the server, so it cannot be bypassed from the client side.
3	Database / Data Storage	ServiceNow Platform Database (relational tables such as Incident, User [sys_user], Group, Role)	Built-in relational storage with role, group and user tables that ACLs can check. Table and field-level security is native, and no separate database setup is required.
4	Cloud / Hosting / Deployment	ServiceNow Personal Developer Instance (PDI), hosted on the ServiceNow Cloud	Free, fully managed cloud instance with no infrastructure to maintain. Update Sets move the configuration between instances. Scalable and secure by default.
5	Version Control & CI/CD	ServiceNow Update Sets and Source Control integration (GitHub)	Tracks and packages ACL, script and role changes, and supports collaboration and rollback. Update Sets deploy the tested changes from Dev to Test/Prod.
6	Third-Party APIs / Other Tools	ServiceNow Script Debugger, Impersonate User, Access Analyzer / Security Diagnostics, ServiceNow REST Table API	Impersonate User and Access Analyzer verify that the ACL restricts records correctly for each role. Script Debugger helps debug the ACL scripts. The REST Table API is used to test that the access restriction also holds for API calls.